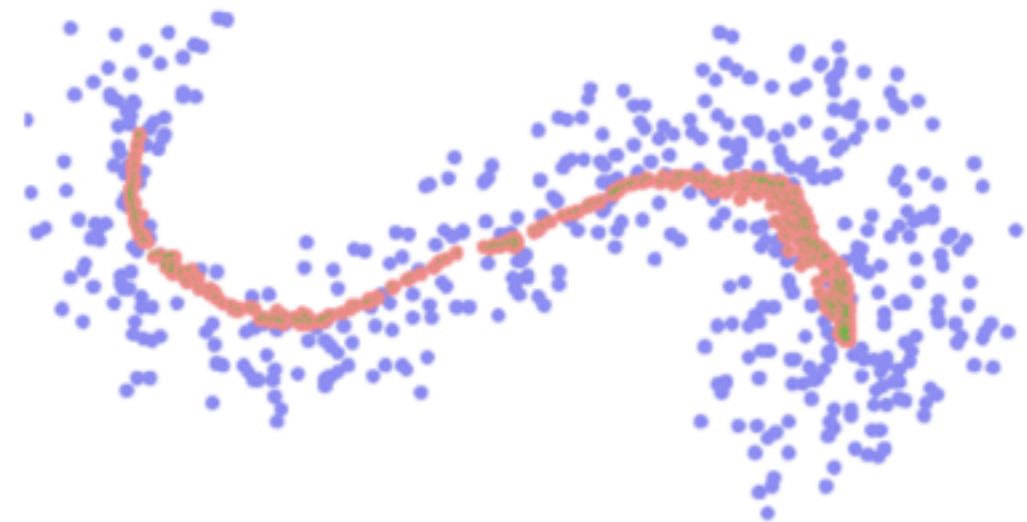


$$\min_{\alpha} W_{\varepsilon,p}^p(\alpha, \beta)$$



$$\min_{\alpha} \overline{W}_{\varepsilon,p}^p(\alpha, \beta)$$

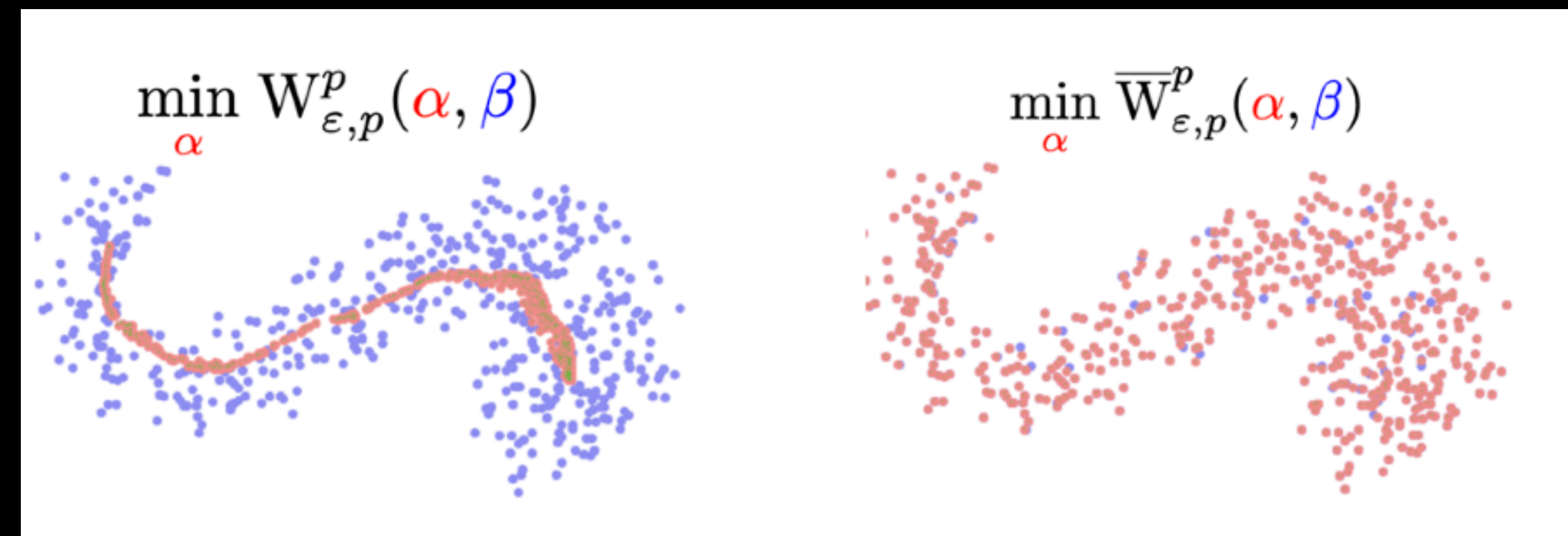


Sinkhorn Divergence

- Debiased definition [Ramdas et al. 2017; Genevay et al. 2018]:

$$S_\varepsilon(\alpha, \beta) = \text{OT}_\varepsilon(\alpha, \beta) - \frac{1}{2}\text{OT}_\varepsilon(\alpha, \alpha) - \frac{1}{2}\text{OT}_\varepsilon(\beta, \beta)$$

- Known as the Sinkhorn Divergence
- Properties:
 - Symmetric, nonnegative, definite.
 - Interpolates between: $\text{MMD} (\varepsilon \rightarrow \infty) \leftrightarrow \text{OT} (\varepsilon \rightarrow 0)$.
- Metric-like, stable in ML practice.



Application: SD for Generative Modeling

Learning Generative Models with Sinkhorn Divergences

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Gabriel Peyré
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École Normale Supérieure

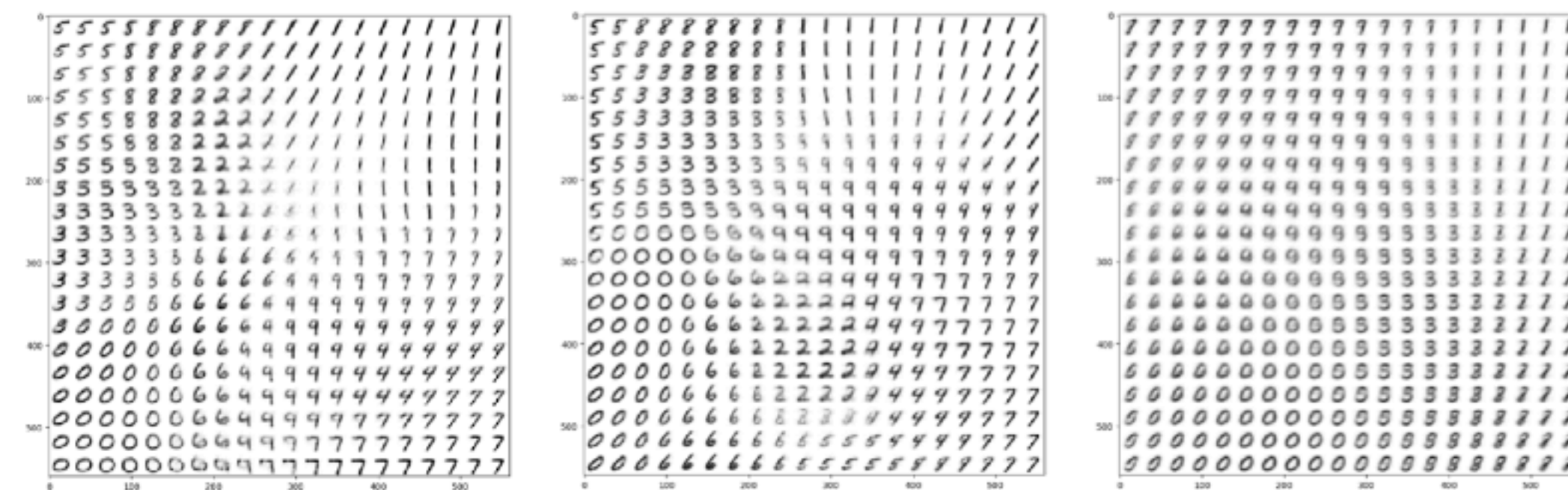
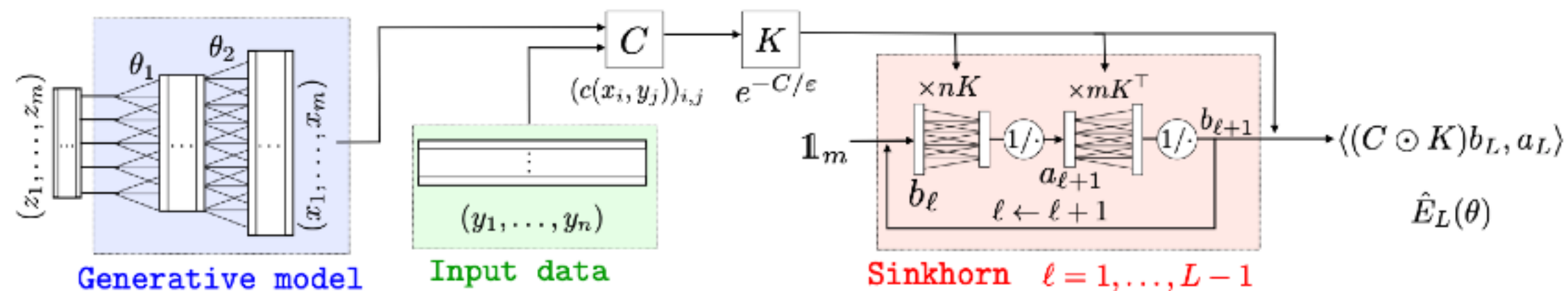
Marco Cuturi
CREST ENSAE
Université Paris-Saclay

Abstract

The ability to compare two degenerate probability distributions, that is two distributions supported on low-dimensional manifolds in much higher-dimensional spaces, is a crucial

1 Introduction

Several important statistical problems boil down to fitting densities, *i.e.* estimating the parameters of a chosen model that *fits* observed data in some meaningful way. While the standard approach is maximum likelihood estimation, this approach is often flawed in

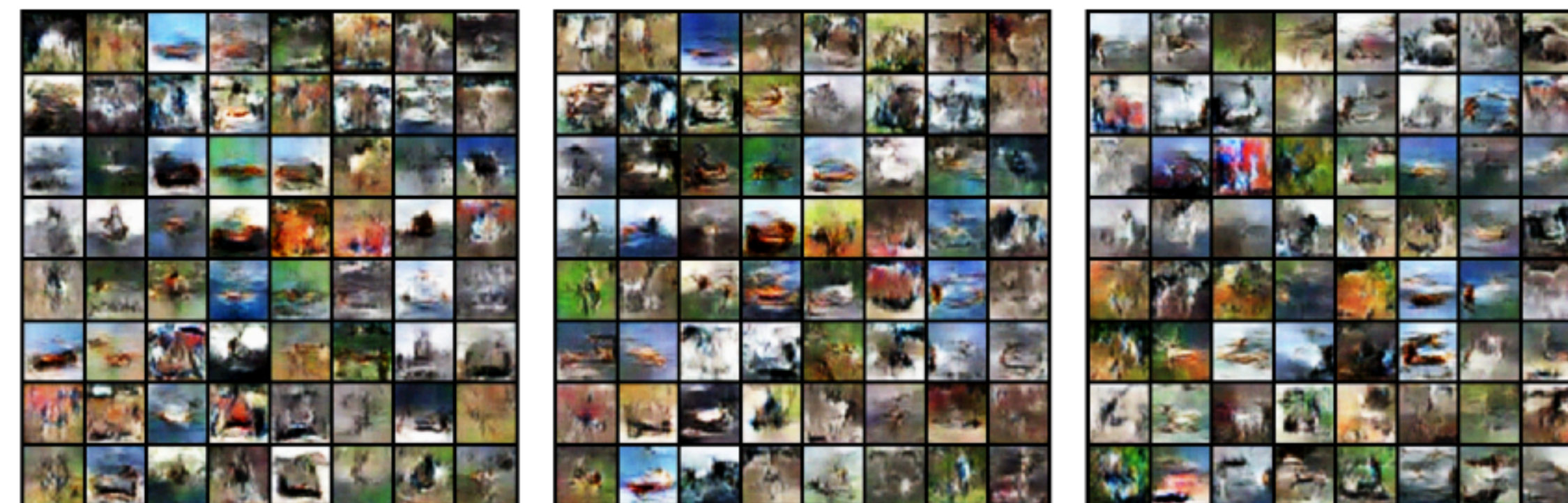


(a) $\epsilon = 1$
 $m = 200, L = 10$

(b) $\epsilon = 10^{-1}$
 $m = 200, L = 100$

(c) $\epsilon = 10^{-1}$
 $m = 10, L = 300$

Figure 3: Influence of the hyperparameters on the manifold of generated digits.



(a) MMD

(b) $\epsilon = 1000$

(c) $\epsilon = 10$